AgriGuard – SFIC Portal Answers

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Copy each answer into the matching step on the SFIC portal. Every answer is within its word limit (count shown in brackets), and the concept PDF for Step 18 is ready.

Steps 1–5: Basic details

Step	Field	Enter this
1	Title of the Solution	AgriGuard – Early Crop Disease Screening and Safe Advisory for Small Farmers
2	Theme	Samriddh Annadata, Samriddh Bharat
3	Sub-theme	Agricultural Innovation (category: Undergraduate and Above)
4	Problem Statement	Small and medium farmers in India cannot identify crop diseases and pest attacks early or get timely, reliable advice, so outbreaks spread unchecked and destroy a large share of their harvest and income.
5	Track	Track 1: Product / Technology Solution

Step 6: Describe the Problem (373 / 500 words)

Crop diseases and pests are one of the biggest causes of income loss for Indian farmers. An ICAR official has stated that about 30–35% of India's annual crop yield is lost to pests. The problem is not only the pest itself but late detection: by the time a farmer is sure what is wrong, the damage has spread across the field.

Most small farmers identify problems by eye, by asking neighbours or by asking the local pesticide shop. Expert help is scarce. India has fewer than one agricultural extension worker for every 5,000 farmers, against a national norm of about 1:750 in rainfed and 1:1,100

in irrigated areas (NAAS/ICRISAT, 2025). As a result, farmers often guess and spray several chemicals that do not work, adding cost and harming soil, health and the environment.

Recent outbreaks show how often and how widely this happens:

- Cotton, 2023 – pink bollworm destroyed about 65% of the crop in Punjab and Haryana and up to 90% in parts of Rajasthan. Farmers had debts of ₹6–8 lakh, and Rajasthan announced ₹1,125 crore in relief after the crop was lost (Down To Earth).
- Chilli, 2021–22 – black thrips hit at least 3.6 lakh hectares in Telangana and Andhra Pradesh; yields fell from 5–6 t/ha to 2.5 t/ha or less. One farmer used six pesticides and none worked; another family said nobody came to explain the pest (Down To Earth).
- Coconut, Tamil Nadu, 2024–25 – root wilt and whitefly cut yields by 30–40% in the Pollachi belt; 1.9 lakh acres were reported affected and farmers cut down hundreds of trees (DT Next, The Quint).
- Banana, 2020 – Panama wilt (TR4) infected over 11,000 hectares in Bihar and Uttar Pradesh; farmers' earnings fell to less than half (Down To Earth).
- Soybean, Madhya Pradesh, 2025 – a yellow mosaic outbreak led the state to pay ₹322 crore to about 4.95 lakh farmers (news reports).

The problem recurs every season, in every major crop and region. Pink bollworm losses in north India have repeated since 2017. In each case the pattern is the same: late or wrong identification, no timely expert guidance, money wasted on ineffective sprays, and government relief that arrives only after the harvest is lost.

Step 7: Explain Your Proposed Solution (777 / 1,000 words)

AgriGuard is a mobile-first web application that helps a farmer check a sick plant early and decide what to do next. The farmer takes a photo of an affected leaf with an ordinary smartphone (choosing the crop, or letting AgriGuard detect it automatically), and within seconds receives a screening result: whether the leaf looks healthy or shows signs of a likely disease, how confident the system is, an estimate of how much of the leaf is affected, and simple, safe next steps in their own language.

What the farmer gets

- Likely problem, or "Healthy", with a confidence level shown honestly. The AI fully covers 5 crops – tomato, potato, chilli/pepper, maize and grape (23 classes of diseases and pest damage). For other crops it only checks the leaf for visible problems and never guesses a disease.

- A "Not sure" result when the photo is unclear or the system is not confident, with advice to retake the photo or contact the local agriculture officer or Krishi Vigyan Kendra (KVK).
- A severity estimate (Low / Moderate / High) based on the visible affected area of the leaf.
- Advisory content: immediate actions, symptoms to confirm, prevention, low-risk organic and cultural practices, soil and irrigation guidance, and when to seek expert help. AgriGuard deliberately does not prescribe pesticide brands or doses; every advisory ends with a referral to the local agriculture officer or KVK, so the tool cannot push farmers into wrong or excessive spraying. An automated test blocks any chemical brand name from entering the advisory.
- Scan history and before/after comparison, so the farmer can see whether a problem is spreading or improving.
- A dashboard showing healthy vs affected scans and trends over time.
- English and Tamil with an instant language switch; other Indian languages planned.
- Personal accounts so each farmer's scans and photos stay private.

How it works (summary)

1. Photo check: blurry, dark or non-leaf images are rejected with a "retake" message.
2. Crop-aware classification: a deep-learning image model (transfer learning from a public, licence-checked crop-disease dataset) identifies the crop, then predicts only among problems of that crop, which reduces confusion.
3. Confidence gate: high confidence shows the result; medium shows "possible" with the top three options; low shows "Not sure" and an expert referral.
4. Severity: image processing measures the share of the leaf showing lesions.
5. Advisory lookup: reviewed content, stored as data so agriculture experts can update it without changing code.
6. Results are saved to the farmer's history for comparison.

Optional field sensing

A low-cost ESP32 kit (about ₹1,000) with soil-moisture, temperature and humidity sensors sends field readings to the AgriGuard app and dashboard. These readings are shown as context (for example, humid and wet conditions favour fungal disease), not as a diagnosis.

SMS alerts for button phones

A low-cost GSM module (SIM800L, about ₹500) can be added to the kit. When the soil is too dry or conditions favour disease, the ESP32 sends a short SMS – for example, "AgriGuard: high humidity for 2 days – check leaves for fungal spots" – to any mobile number. Farmers with basic button phones and no internet therefore also get alerts. Where Wi-Fi or a phone hotspot is available, the same SMS can be sent through an internet SMS service with no extra hardware.

Technology

Next.js progressive web app (works in a phone browser, no app-store install; an Android wrapper app is also built), Python FastAPI backend, PyTorch model behind a replaceable adapter so a better or field-trained model can be swapped in by configuration, SQL database (SQLite on a laptop server, PostgreSQL in the cloud) and private image storage per user.

Responsible design

AgriGuard is a screening and decision-support tool, not a replacement for experts. Every result is labelled as an estimate, low-confidence cases are escalated to humans, no chemical prescriptions are given, and farmers' images are used only for their own history unless they agree otherwise.

Current development stage

AgriGuard is at the prototype stage. The full application is built and runs end to end: accounts, photo upload with quality checks, result page with confidence and severity, English and Tamil advisory, history, before/after comparison, dashboard, and ESP32 sensor input with a simulator. 40 automated backend tests pass, covering photo quality, confidence rules, severity, the no-chemical-name rule, the KVK referral and the AI model. A trained image model now runs on the laptop server and inside the phone browser: 94.7% correct on lab photos and 61.5% on real field photos. The next steps are to collect local field photos to improve accuracy, have experts review the advisory, and run a small pilot with farmers and a KVK.

Step 8: Explain Who Is Affected (178 / 300 words)

- Small and marginal farmers – the majority of Indian farmers, who have little access to experts and depend on one or two crops for their yearly income. Many farm leased land with borrowed money, so one outbreak can push a family into debt.

- Farmers growing high-value, disease-prone crops – cotton (Punjab, Haryana, Rajasthan, Telangana), chilli (Telangana, Andhra Pradesh, Karnataka, Tamil Nadu), coconut (Coimbatore and Tirupur in Tamil Nadu, Karnataka, Kerala), banana (Bihar, Uttar Pradesh, Tamil Nadu), soybean (Madhya Pradesh), and vegetables such as tomato and potato.
- Farm labourers, whose work and wages fall when crops fail.
- Rural households and villages that depend on farm income, including women farmers who often manage fields with even less access to extension services.
- Extension officers and KVK staff, who cannot personally reach thousands of farmers each season.
- State governments, which pay large relief amounts after outbreaks (for example ₹1,125 crore in Rajasthan for cotton in 2023 and ₹322 crore in Madhya Pradesh for soybean in 2025).
- Consumers, who face price rises when outbreaks reduce supply.

Step 9: Highlight Innovation (225 / 300 words)

Existing tools such as Plantix, the Kisan Call Centre and the National Pest Surveillance System either always return an answer, depend on an expert being available, or are linked to chemical sales. AgriGuard is a safety-first, low-cost screening tool:

- Honest "Not sure" instead of a forced answer: AgriGuard shows its confidence and refuses to guess when unsure, sending the farmer to an expert. In field tests only 2 of 28 healthy leaves were called diseased.
- Photo + field sensors: an optional ₹1,000 ESP32 kit checks soil moisture, temperature and humidity, warning when conditions favour disease – not only after symptoms appear. An optional GSM module sends these warnings by SMS, so they reach basic button phones too.
- Severity, not just a label: it estimates how much of the leaf is affected (Low / Moderate / High) and tracks change over time with before/after comparison.
- No chemical selling: no pesticide brands or doses, enforced by an automated test; every result refers serious cases to the KVK.
- Leaf-only check: soil, pots and dark backgrounds are ignored, so they are never mistaken for disease.
- Built for rural India: works in a phone browser with no app install, Tamil first, and the AI runs on the phone itself.

- Works with the government system (KVK, Kisan Call Centre, NPSS) rather than replacing it.

Step 10: Explain How the Solution Works (227 / 300 words)

1. Open: the farmer opens AgriGuard in a phone browser and chooses a language.
2. Select crop: the farmer picks the crop, or chooses auto-detect.
3. Capture: the farmer photographs one affected leaf in good light.
4. Quality check: the system rejects blurry, dark or non-leaf photos and asks for a retake.
5. Screening: the image is compressed and passed to a deep-learning model that identifies the crop and predicts only among that crop's problems.
6. Confidence gate: high confidence shows the likely disease; medium shows "possible" with the top three options; low shows "Not sure" with advice to consult the local agriculture officer or KVK.
7. Severity: image processing separates the leaf from the background and measures the share of diseased area, giving Low, Moderate or High.
8. Advisory: the system shows reviewed guidance on symptoms, prevention, sanitation and when to seek help, with no chemical prescriptions.
9. Save and track: the result is stored in the farmer's history, which can be compared over time; the dashboard shows trends.
10. Optional sensors: an ESP32 kit sends soil moisture, temperature and humidity every minute to show field conditions alongside scans; with the optional GSM module it also sends SMS alerts to the farmer's phone, even a button phone.
11. Improve: field photos (with consent) and expert feedback are used to retrain and update the model and advisory.

Step 11: Development Stage

Select: Prototype

Additional Notes on Your Stage (177 / 300 words)

The solution is at the prototype stage. Completed so far: the problem was studied using recent Indian outbreak reports and national data. The mobile-first web app is built with accounts, crop selection or auto-detect, photo upload with quality checks, result page with confidence and severity, English and Tamil advisory for the first 10 disease classes (tomato, potato, pepper, cucumber), scan history, before/after comparison and

dashboard. The FastAPI backend runs the full scan flow end to end. ESP32 firmware and a sensor simulator feed field conditions to the dashboard, and an Android wrapper app exists. A trained AI model (94.7% correct on lab photos, 61.5% on real field photos) runs on the server and in the phone browser, and 40 automated backend tests pass.

Remaining work: local field photos are needed to raise field accuracy and add crops such as paddy, banana and coconut; a farmer survey is planned; advisory content needs expert review and extension to more crops; and testing on low-end phones and a small pilot with farmers and a KVK are planned before wider use.

Step 12: Expected Impact (196 / 300 words)

Economic: earlier detection lets farmers act while damage is still limited, protecting yield and income. Even a small reduction in the 30–35% of yield lost to pests is significant for a family whose income depends on one season. Avoiding wrong or repeated pesticide sprays also cuts input costs; in the chilli outbreak one farmer used six pesticides without success.

Social: farmers get guidance on the same day instead of waiting for an officer or relying on shopkeepers. This reduces the stress and debt that follow crop failure, which in recent outbreaks has been linked to farmer suicides. Local-language support and a simple design include farmers with limited literacy and women farmers, and SMS alerts from the field kit reach farmers who only have basic button phones.

Environmental: no-prescription advisory and early sanitation steps reduce unnecessary chemical use, protecting soil, water, pollinators and farmworkers' health.

Institutional: KVKs and agriculture departments can reach many more farmers, and anonymised, consented scan data can show where a disease is appearing, supporting earlier warnings and better-targeted relief instead of relief only after losses.

Educational: farmers learn to recognise symptoms and good field practices over time through the advisory and history features.

Step 13: Estimated Cost (283 / 300 words)

The AgriGuard field kit costs about ₹1,000 to build. The software costs nothing, because it uses open-source tools and free cloud tiers.

Hardware – one AgriGuard field kit (approx.)

- ESP32 microcontroller (Wi-Fi): ₹600

- Soil moisture sensor: ₹200
- Humidity and temperature sensor: ₹150
- Wires, soldering and simple casing: ₹50
- Total per kit: about ₹1,000

To keep the cost low, the kit has no separate display: readings are shown in the AgriGuard app on the farmer's phone. An LCD display can be added for about ₹200 where needed.

Optional SMS add-on – for farmers with button phones (approx.)

- SIM800L GSM module (5 V board with antenna): ₹450
- 1000 µF capacitor and extra wires: ₹50
- Total add-on: about ₹500, so a kit with SMS alerts costs about ₹1,500
- Running cost: a prepaid 2G SIM (BSNL, Airtel or Vi) with an SMS pack, a small monthly recharge per kit. The module runs from a 5 V phone charger, so no extra battery is needed.

Free alternative: where Wi-Fi or a phone hotspot is available, the ESP32 can send the same SMS through an internet SMS service, with no extra hardware.

Software – ₹0: the app, AI model and backend are built with open-source tools (Next.js, FastAPI, ONNX Runtime) and open crop-disease datasets; hosting runs on free tiers at prototype scale.

For farmers: the photo-screening app is free to use on their own smartphone. The sensor kit is optional, at about ₹1,000 per field.

At pilot scale (50–100 farmers), the main extra costs are around 10 sensor kits (about ₹10,000, or about ₹15,000 with SMS modules) plus field-photo collection and travel. These can be covered by a KVK, FPO, agriculture department or CSR partner rather than by farmers.

Step 14: Implementation and Scalability (193 / 300 words)

Phase 1 – Pilot (3–6 months): finish the trained model for 3–5 common local crops, test with real field photos, and run a pilot with 50–100 farmers through a nearby Krishi Vigyan Kendra or farmer producer organisation (FPO), with agriculture experts reviewing results.

Phase 2 – District scale: add more crops and diseases based on pilot demand, complete Tamil and add other regional languages, and train extension officers and FPO volunteers to onboard farmers. Expert feedback and consented field photos are used to improve

accuracy.

Phase 3 – State and national scale: partner with state agriculture departments and agricultural universities to add region-specific crops and advisory content. Because AgriGuard runs in a phone browser, no app-store install or special hardware is needed, and cloud hosting scales with users.

Scalability features already in the design: replaceable model, advisory stored as editable data, multi-language structure, and low-cost optional sensors. Future additions include an offline mode for low-connectivity villages, voice output for low-literacy users, WhatsApp summaries (SMS alerts through the GSM add-on are already part of the kit design), and anonymised disease maps that alert officers when many scans show the same disease in one area.

Step 15: Provide Evidence

Select: Prototype, Other

Prototype: a working app and backend that runs the complete scan flow (crop selection, photo upload, result with confidence and severity, English/Tamil advisory, history and dashboard) with a trained AI model (94.7% lab, 61.5% field accuracy); 40 automated backend tests pass; ESP32 sensor firmware and simulator included.

Other: a case-study document compiled from published Indian sources – Down To Earth, DT Next, The Quint, ICAR statements and NAAS/ICRISAT data – documenting five recent outbreaks (cotton, chilli, coconut, banana, soybean) and national figures on crop loss and the shortage of extension workers. Field survey and accuracy testing are planned for the pilot stage.

Step 16: Identify Potential Adopters (133 / 300 words)

- Krishi Vigyan Kendras (KVKs) under ICAR – to give farmers quicker first-level screening and refer serious cases to scientists.
- State agriculture and horticulture departments, including the Tamil Nadu Department of Agriculture and its extension officers.
- Agricultural universities and research institutes such as Tamil Nadu Agricultural University (TNAU) and ICAR institutes – for validating advisory content and supplying expert-labelled images.

- Farmer Producer Organisations (FPOs) and cooperatives – to onboard member farmers and use disease trends for planning.
- Agricultural Technology Management Agencies (ATMA) and Common Service Centres – for village-level reach.
- NABARD and rural development programmes – for funding and integration with farmer-support schemes.
- Agri-focused NGOs and CSR foundations working with small farmers.
- Digital agriculture platforms run by government, where AgriGuard could serve as a screening module.

Step 17: Mention the Support Required (139 / 300 words)

- Mentorship from plant pathologists and agriculture scientists (KVK, agricultural university or ICAR) to review disease classes and advisory content and guide safe recommendations.
- Access to expert-labelled field images of Indian crops, and permission to collect photos on farms through a KVK or FPO.
- Technical mentorship on deploying deep-learning models efficiently on low-end phones and slow networks, including an offline mode.
- Cloud computing credits for model training, hosting and image storage during the pilot.
- Seed funding of about ₹20,000–50,000 for field data collection, sensor kits, travel and translation review.
- A pilot partner (KVK, FPO or state agriculture department) to test AgriGuard with 50–100 farmers and give structured feedback.
- Support with language translation and voice content in Tamil and other regional languages.
- Guidance on data privacy, intellectual property and the path to incubation.

Steps 18–21: Uploads and final submission

- [x] ~~**Step 18 – Concept document (mandatory):** upload AgriGuard_SFIC_Concept_Document.pdf (PDF, 8 pages, 0.9 MB; limit 15 MB) – the latest version with the ₹1,000 kit and the existing-solutions comparison~~

- [ ] **Step 19 – Supporting documents (optional, up to 50 MB):** export the case-study doc (Crop Disease & Pest Losses in India) as PDF and upload it; add your own farm photos if taken
- [ ] **Step 20 – Video (optional):** record a 1–2 minute screen video of a scan from photo to result, upload to YouTube (unlisted) or Google Drive, paste the link
- [ ] Read every field once more and check team details on the portal
- [ ] **Step 21:** click SAVE AS DRAFT first, then SUBMIT IDEA before 24 October 2026

Portal: sevafirst.mic.gov.in